Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	HEV-Activated Photochromic Dyes and Optical Film with Rapid Indoor Fade-Back for Display			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Fine Chemicals	Dyes/Pigments & Intermediates	Display	Display Components and Materials
Concept of Proposed Technology	**Concept of Proposed Technology** This project targets photochromic dye chemistry and optical film architectures optimized for transparent or/and wearable displays, for augmented reality (AR) and smart glasses. The key differentiation is (i) activation under high-energy visible (HEV) blue-violet light (e.g., 400–430 nm) and (ii) rapid fade-back under indoor visible light (>450 nm), enabling high transmittance and low color distortion when the display is used indoors [1,4,8]. **Background and unmet needs** • Conventional UV-activated photochromism shows limited activation behind UV-filtering glass (e.g., no activation behind automotive windshields) and may also not meet the fast, display-compatible fade-back needed for AR/transparent displays [2,3].			

	• Display integration requires high clarity (high luminous transmittance, low haze), controlled color shift, and durability against repeated switching (fatigue) and environmental stress (temperature, humidity, UV/visible cycling) [8–13]. **Proposed technological development** • Mechanism-guided molecular design to tune absorption and switching kinetics, leveraging visible-light-triggered photochromic design principles and established photochromic families (e.g., naphthopyrans, spiropyrans, diarylethenes) [4–8,14]. • Film and stack engineering (polymer matrix selection, dye–polymer interaction control, barrier/protection layers) to meet optical and reliability requirements for display lamination. [8–13] • Quantitative benchmarking against representative "in-car" photochromic products that are designed to activate behind the windshield (e.g., Transitions XTRActive; HOYA Sensity) to set competitive performance targets [2,4]. **Target applications** • AR smart glasses / transparent displays: outdoor glare reduction and rapid indoor recovery for display readability. • Automotive heads-up display (HUD) / smart glazing and other display-adjacent optical components requiring fast switching and low haze.
Rationale for Joint R&D Project	**Rationale for International Joint R&D Project** The project requires an end-to-end capability chain from molecular switch design to scalable synthesis, optical film formulation, and display-grade reliability validation. This is difficult [8,14] to achieve within a single organization and therefore benefits from an international joint R&D structure. **Complementary roles** • Overseas research institute: photochemical/photophysical mechanism study, computational and synthetic design of HEV-activated / fast-bleaching dye candidates, and small-scale synthesis protocols [4–8]. • Korean SME partner: scalable synthesis and purification, polymer/film formulation & fabrication, lamination-compatible film stack design, and device-level optical/reliability testing for smart glasses and transparent display environments [8–13]. • IP, data exchange, and commercialization planning will be prepared to meet program requirements and to support the Korean SME's global market entry.
Key Words	1. HEV-activated photochromic dyes 2. automotive HUD 3. Low-haze optical film 4. transparent display 5. visible-light-triggered photoswitch

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 5
R&D Objectives	• Elucidate the switching mechanism under HEV/visible excitation and identify molecular design rules for HEV activation and rapid indoor fade-back. • Design and synthesize a small library of candidate dyes (lead + backup candidates) with improved kinetics and reduced residual tint, guided by structure–property relationships in established photochromic systems. • Improve fatigue resistance and environmental stability (thermal, UV/visible cycling, humidity) through combined molecular and matrix (film) engineering. • Establish scalable synthesis/purification routes and quality specifications suitable for film production (batch reproducibility and impurity control). • Demonstrate low-haze, high-transmittance optical film prototypes and validate performance using standardized optical test methods, benchmarking against representative commercial in-car photochromic products.			
R&D Target Specifications	• Activation band: switching response in the HEV (blue-violet) band; target peak response 400–430 nm [2,4]. • Indoor fade-back: half-life (t1/2) ≤ 2 min at 23 °C under defined indoor white-LED lighting; residual tint after 5 min ≤ 10% of outdoor activated OD [4,8]. • Clear-state transparency (film): total luminous transmittance ≥ 90% measured per ISO 13468-1 (or equivalent) [10] • Haze (film): ≤ 1.0% (goal ≤ 0.5%) measured per ISO 14782 or ASTM D1003 [11,12]. • Color shift: $\Delta E \times ab$ (CIELAB) ≤ 2 in the clear state; measured/calculated per ISO/CIE 11664-4 (or equivalent) [14]. • Radiation durability (fatigue): After high-energy light exposure, ≤ 10% loss of optical density and ≤ 20% increase in fade-back half-life [9]. • Eyewear/transmittance compliance (if applicable): transmittance requirements will be checked against ISO 8980-3 for spectacle-lens use cases [9].			
R&D Plan	**Stage 1 (0–6 months): Pre-research (planning + feasibility).** Detailed R&D planning • Overseas RI: switching mechanism study, HEV-activation strategy, candidate shortlist, and small-scale synthesis protocols. • Korean SME: manufacturability/cost screen, polymer compatibility study, early film formulation tests, and set-up of optical measurement methods (transmittance, haze, color). • Joint: down-select lead candidates, lock KPIs and test protocols, finalize Stage 2 work packages, and prepare the market/IP strategy and collaboration governance package. • Evidence package: preliminary kinetics & optical KPI results, risk register, and a submission-ready. **Stage 2 (7–36 months): Main research.** Prototype development and validation • Korean SME: scale-up synthesis/purification, film stack engineering (barrier/adhesion), lamination process development, and prototype film/module production with QA. • Overseas RI: continued molecular optimization support, advanced spectroscopy/kinetics analysis, and design iterations based on structure–property relationships. • Joint: system-level evaluation and benchmarking vs commercial in-car photochromic products, cycling durability and environmental reliability testing, and delivery of TRL-5 prototypes plus an initial commercialization package. **Project governance & data/IP management** • Monthly technical meetings, quarterly steering committee, shared KPI dashboard, sample/data exchange under MTA/NDA, and joint invention disclosure/patent strategy compatible with program guidance.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Photochemical/photophysical mechanism study for HEV activation and rapid indoor fade-back (spectroscopy, kinetics, computational modeling). • Molecular design and synthesis of candidate dye library (lead + backup candidates), including structure–property mapping and recommendations for film formulation. • Provide small-scale samples, analytical methods, and a transfer package to support scale-up by the Korean SME.
Expected R&R of Korean SME Partner	• Process development for scalable synthesis, purification, and QA specifications. • Polymer/film formulation and multilayer stack design compatible with display lamination (adhesion, barrier, scratch resistance). • Prototype manufacturing and evaluation: transmittance, haze, and color shift (ISO 13468-1; ISO 14782 / ASTM D1003; ISO/CIE 11664-4) plus cycling/environmental durability tests. [9–13] • Commercialization: customer discovery in the smart-glasses / transparent-display value chain, supply-chain planning, and manufacturing readiness.

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical outcomes** • Mechanism and design rules for HEV-activated / fast-bleaching photochromic dyes suitable for display applications, supported by a characterized candidate dye set and dataset. • TRL-5 prototype optical film / laminated coupon demonstrating high transmittance, low haze, fast indoor fade-back, and improved fatigue resistance vs benchmark products. • Standardized KPI and test protocol package (transmittance, haze, color, cycling) enabling repeatable evaluation and future certification work. **IP and commercialization outcomes** • Joint invention disclosures and patent filings (molecules, formulations, film stack, and use cases), plus an FTO and licensing strategy; structured to align with program IP guidance. • Korean SME-led path to enter global smart-glasses / transparent-display supply chains, including pilot customer evaluations and scale-up planning. **Impact** • Improve the SME's global competitiveness and overseas market entry readiness by generating standardized performance data, prototype samples, and an industrializable manufacturing route suitable for collaboration with global smart-glasses / AR / transparent-display OEM/ODM ecosystems. • Contribute to the creation of a new high-value materials supply opportunity in the Semiconductor / Display strategic field, enabling improved usability and comfort of wearable displays by managing glare while maintaining indoor readability. • Drive measurable economic ripple effects through (a) new product lines and B2B supply contracts, (b) increased export feasibility, and (c) expansion of skilled jobs in materials synthesis, film process engineering, and optical characterization: impact elements emphasized in the program's "business feasibility/economic effect" evaluation items. • Establish a repeatable international collaboration model (regular technical workshops/seminars, exchange of samples/data, and joint planning) that can evolve into mid-to-long-term cooperation and a sustainable partnership network for future global projects.

6. References

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(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization